DG Interns Hub

Project Title:

Final Cybersecurity Virtual Lab – Multi-Service Deployment & Hardening

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Internship:

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Acknowledgement

I would like to express my sincere gratitude to **DG Interns Hub** for providing me the opportunity to work on this cybersecurity internship project.

This project has enhanced my practical understanding of cybersecurity concepts, server deployment, and system hardening. I am thankful to the entire training team for creating a structured and hands-on learning experience that helped me gain real-world exposure to network security and service configuration.

Introduction

Cybersecurity is one of the most crucial aspects of modern IT infrastructure, as organizations increasingly rely on digital services for communication, data storage, and business operations. To understand these concepts practically, this project focuses on building a secure virtual lab environment that includes the deployment of multiple network services and applying essential security hardening measures.

The main objective of this project is to create a controlled environment that simulates real-world enterprise systems such as Web Server, DNS Server, Mail Server, and FTP Server, and to secure them using industry-standard techniques like firewalls, access control, SSH hardening, Fail2Ban intrusion prevention, and secure configurations.

Through this project, I learned how servers interact in a network, how services are installed and configured, how to troubleshoot network issues, and how to strengthen systems against common cyber-attacks. This lab also helped me understand the role of network topology design, secure communication protocols, and system monitoring in maintaining a reliable and secure IT infrastructure.

Purpose of the Project

The purpose of this project is to:

- Build a fully functional **multi-service network environment** using VirtualBox.
- Deploy essential services such as the Web Server, DNS Server, Mail Server, and FTP Server.
- Understand how these services work at the network and system level.
- Apply **security best practices** including firewall rules, port management, SSH modification, user restrictions, and fail2ban.
- Analyze server behavior using tools like **Nmap**, **nslookup**, **dig**, and system logs.

Services Deployed in This Project

The following services were installed, configured, tested, and secured:

1. **Web Server (Apache2 + HTTPS/SSL)**
   Used for hosting web content and securing communication with SSL certificates.
2. **DNS Server (Bind9)**
   Configured for domain name resolution and zone management.
3. **Mail Server (Postfix)**
   Set up to send and receive local mails within the network.
4. **FTP Server (vsftpd)**
   Deployed for secure file transfer with disabled anonymous login and user isolation.

Each service was assigned a specific IP address inside the virtual network:

- **Web Server:** 192.168.56.102
- **DNS Server:** 192.168.56.105
- **Mail Server:** 192.168.56.107
- **FTP Server:** 192.168.56.101

Importance of Cybersecurity & Hardening

Security hardening helps:

- Protect services from unauthorized access

- Prevent brute-force attacks
- Limit network exposure
- Strengthen server configurations
- Improve overall resilience against cyber-attacks

In this project, several key hardening techniques were implemented:

- **UFW firewall** configuration
- **Fail2Ban** to prevent repeated login attacks
- **SSH hardening** (disable root login, change port, allow specific users)
- **FTP security** (disable anonymous login, enable user isolation)
- **SSL encryption** for secure website communication

Lab Setup Overview

This cybersecurity virtual lab was created using Oracle VirtualBox to simulate a real-world network environment consisting of multiple servers and security components. Each virtual machine was configured with essential services, and all systems were interconnected using an internal network to provide isolated and controlled testing.

Virtualization Platform

- VirtualBox Version: Oracle VM VirtualBox 7.x
- Host Operating System: Windows 10 / 11 (64-bit)
- Network Mode Used: *Internal Network (intnet)* for communication between servers
- Optional NAT adapter added for updates

Guest Operating Systems (VMs Used)

VM Role	OS	IP Address	Purpose
Web Server	Kali Linux	192.168.56.102	Host Apache2 and HTTPS

VM Role	OS	IP Address	Purpose
DNS Server	Kali Linux	192.168.56.105	Host Bind9 and zone resolution
Mail Server	Kali Linux	192.168.56.107	Host Postfix for local mail transfer
FTP Server	Ubuntu	192.168.56.101	Host vsftpd for file transfer
Attacker / Test Machine	Kali Linux	DHCP / Manual	Used for Nmap & penetration testing

VM Hardware Configuration

Each Ubuntu server machine was provisioned with:

- **2 GB RAM**
- **2 CPU Cores**
- **20–25 GB Storage**
- **Internal Network Adapter**
- **Optional NAT Adapter for updates**

Tools Installed

Web Server:

- Apache2
- OpenSSL / Self-signed certificate or Let's Encrypt (optional)

DNS Server:

- Bind9
- DNS utilities (dig, nslookup, named-checkconf)

Mail Server:

- Postfix
- mailutils
- Log monitoring tools

FTP Server:

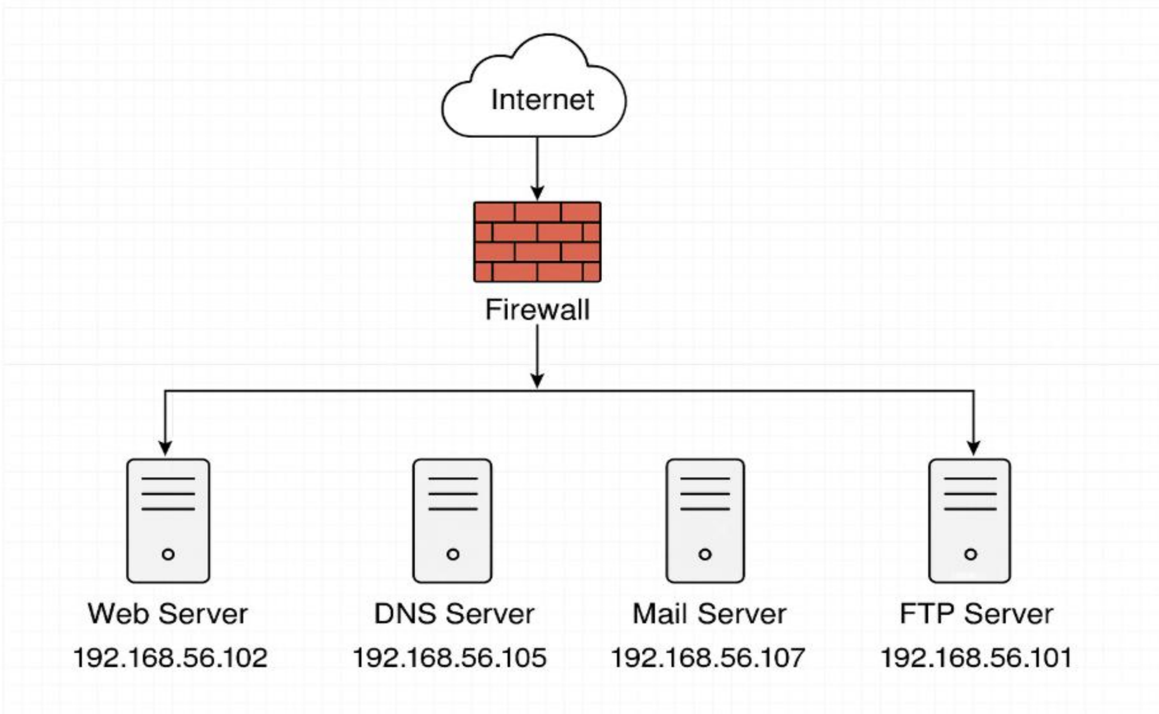
- vsftpd

Security Tools:

- UFW Firewall
- Fail2Ban
- SSH hardening (OpenSSH)
- Nmap (for testing)

Network Topology Diagram

The virtual lab uses an internal network to interconnect all four servers.
Each service has a unique IP address assigned manually.



Description:

- All servers connect to a virtual switch (Internal Network).
- The firewall layer filters traffic.
- Each server runs one primary service.
- Kali Linux performs penetration testing.

Services Configured

1) Web Server (Apache2 + HTTPS)

1.1 **Installation Steps**

```
sudo apt update
sudo apt install apache2
```

1.2 Enable Firewall Rules

```
sudo ufw allow 80/tcp
sudo ufw allow 443/tcp
```

1.3 Create Self-Signed SSL Certificate

```
sudo openssl req -x509 -nodes -days 365 -newkey rsa:2048 \
-keyout /etc/ssl/private/apache.key \
-out /etc/ssl/certs/apache.crt
```

1.4 Configure Virtual Host

Edit the SSL config file:

sudo nano /etc/apache2/sites-available/default-ssl.conf

Set:

SSLEngine on

SSLCertificateFile /etc/ssl/certs/apache.crt

SSLCertificateKeyFile /etc/ssl/private/apache.key

Enable SSL Module:

sudo a2enmod ssl

sudo a2ensite default-ssl.conf

sudo systemctl restart apache2

1.5 HTTPS Testing

Using browser or:

curl -k https://192.168.56.102

Hardening Applied

✓ Disable directory listing

✓ Enable firewall rules

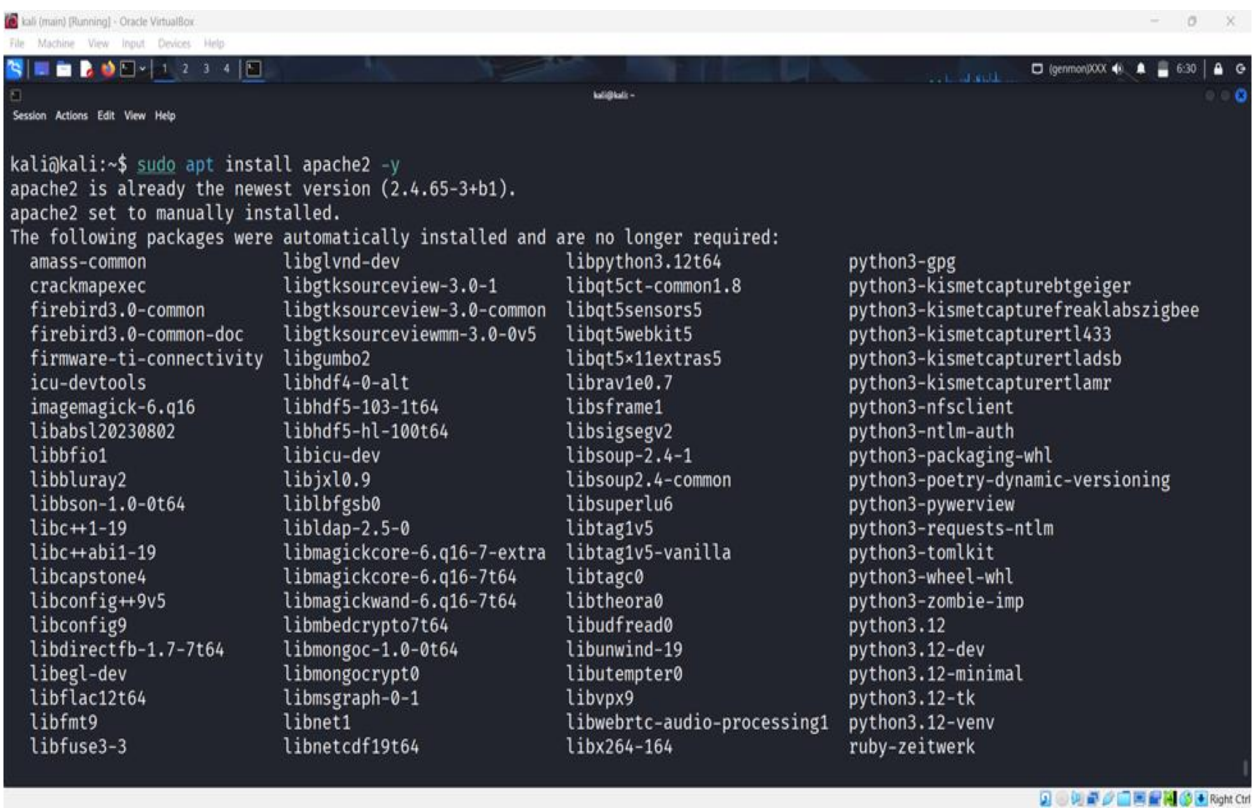
✓ Disable default Apache pages

✓ Restrict access logs

Nmap Scan

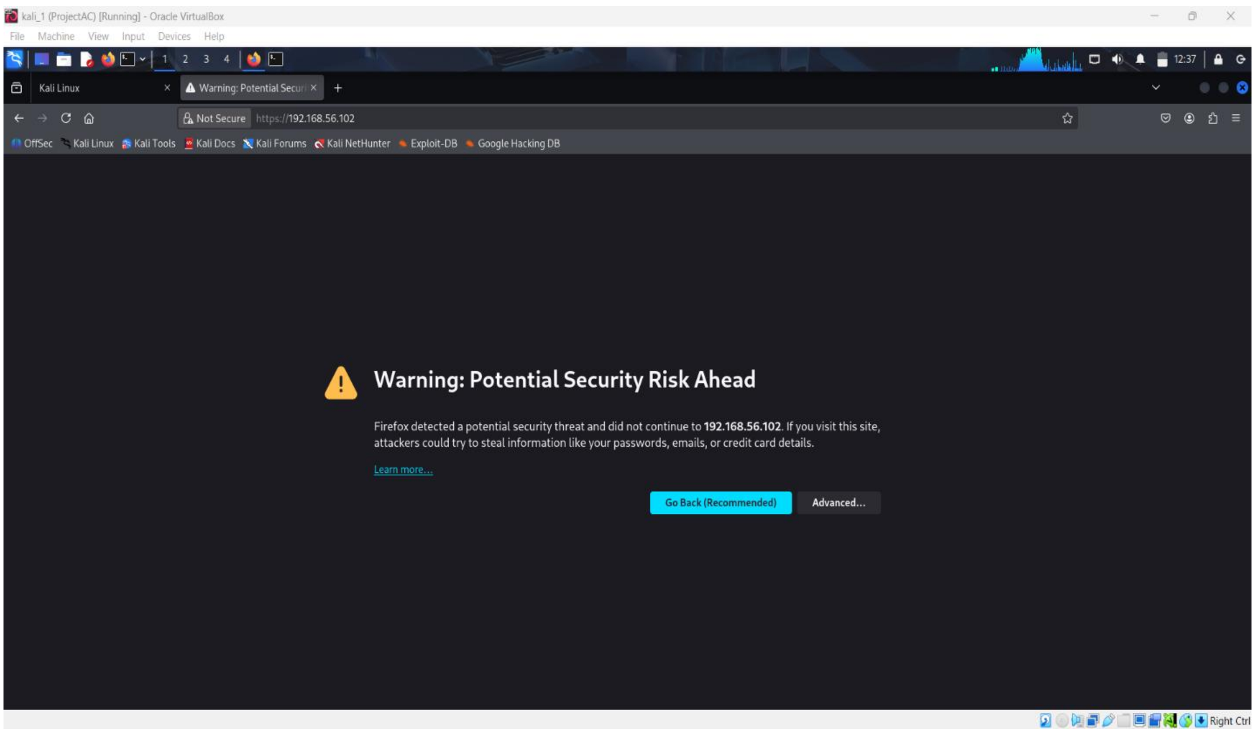
nmap -sV 192.168.56.102

Attach Screenshots:



```
┌──(kali㉿kali)-[~]
└─$ sudo a2enmod ssl
sudo a2ensite default-ssl.conf
sudo systemctl restart apache2

Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Module socache_shmcb already enabled
Module ssl already enabled
Site default-ssl already enabled
```




```
┌──(kali㉿kali)-[~]
└─$ nmap -p 443 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-06 12:39 EST
Nmap scan report for 192.168.56.102
Host is up (0.000078s latency).

PORT    STATE SERVICE
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 0.09 seconds
```

2) DNS Server (Bind9)

2.1 Install Bind9

sudo apt install bind9 bind9utils bind9-doc

2.2 Configure named.conf.local

Add zone:

```
zone "mydomain.local" {
   type master;
   file "/etc/bind/db.mydomain.local";
};
```

2.3 Create Zone File

Path:

/etc/bind/db.mydomain.local

Example zone:

```
@   IN  SOA ns1.mydomain.local. admin.mydomain.local. (
        2 ; Serial
        604800 ; Refresh
```

86400 ; Retry

2419200 ; Expire

604800) ; Negative Cache TTL

;

@ IN NS ns1

ns1 IN A 192.168.56.105

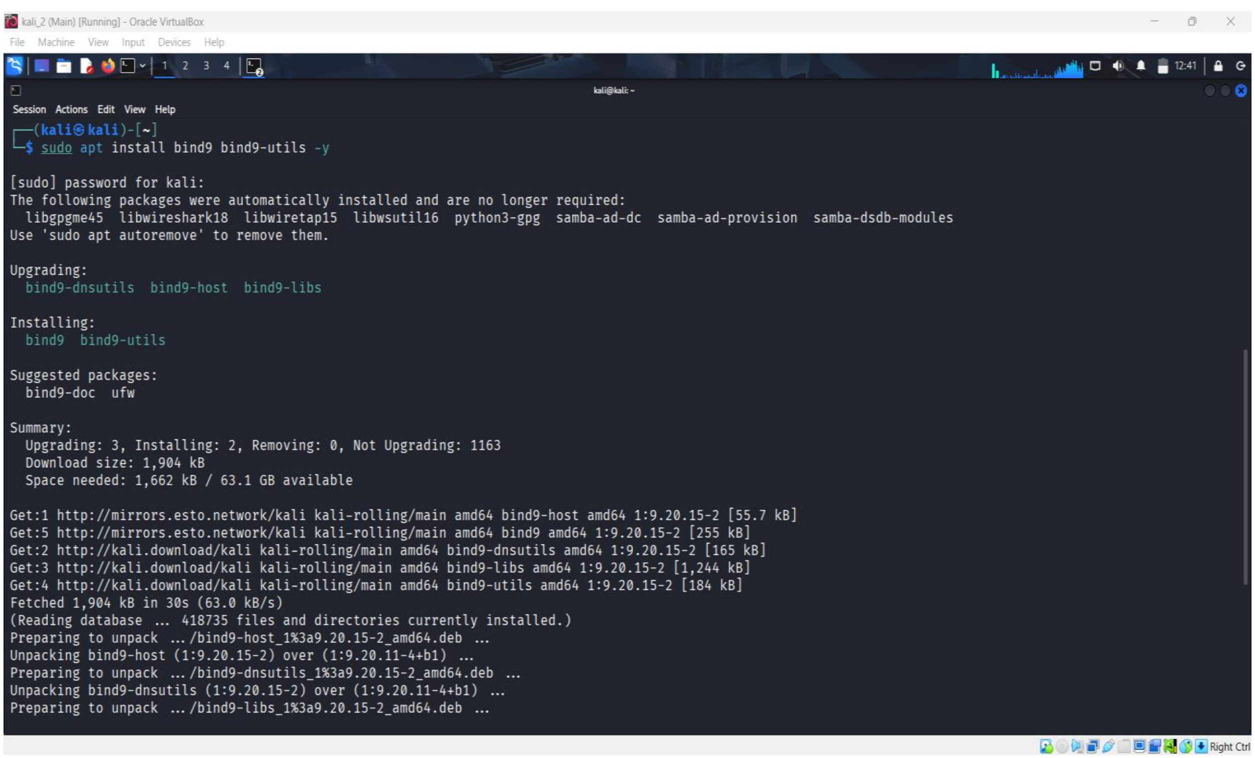
www IN A 192.168.56.102

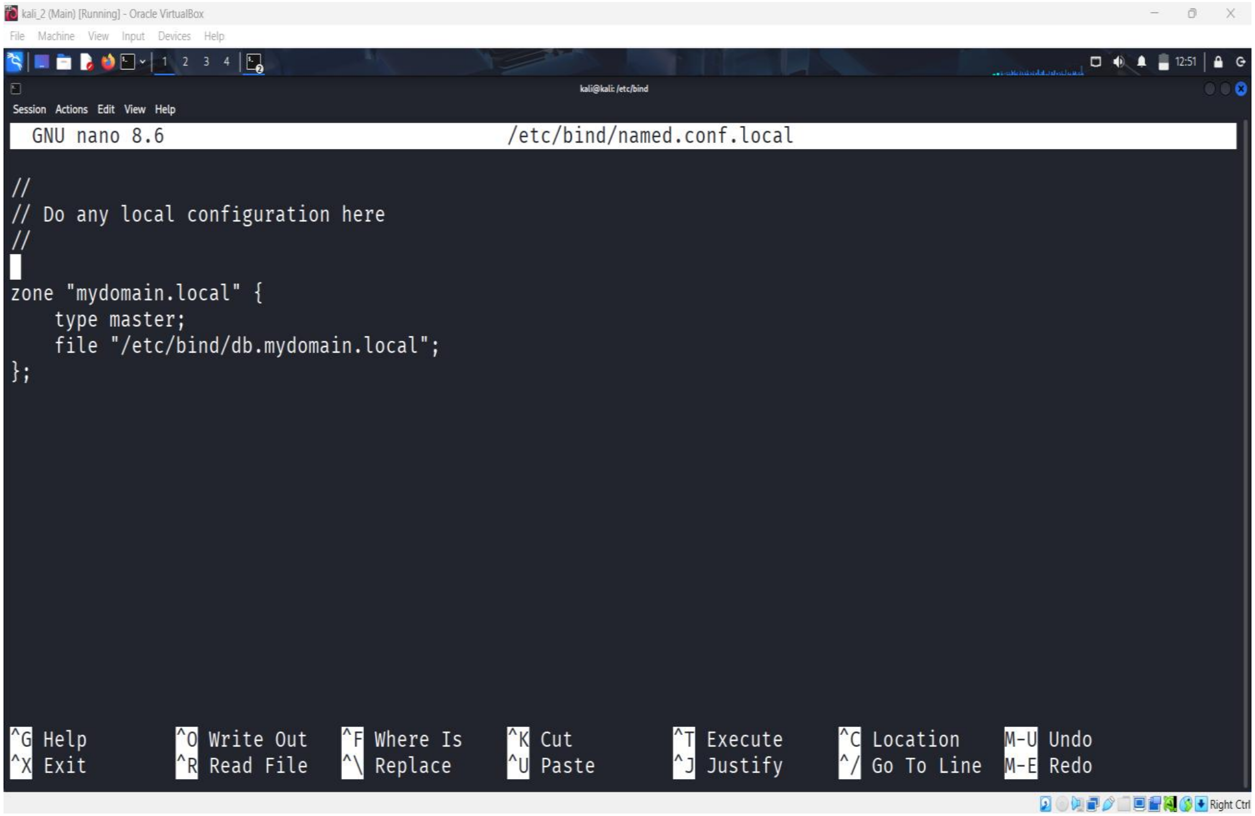
2.4 DNS Resolution Testing

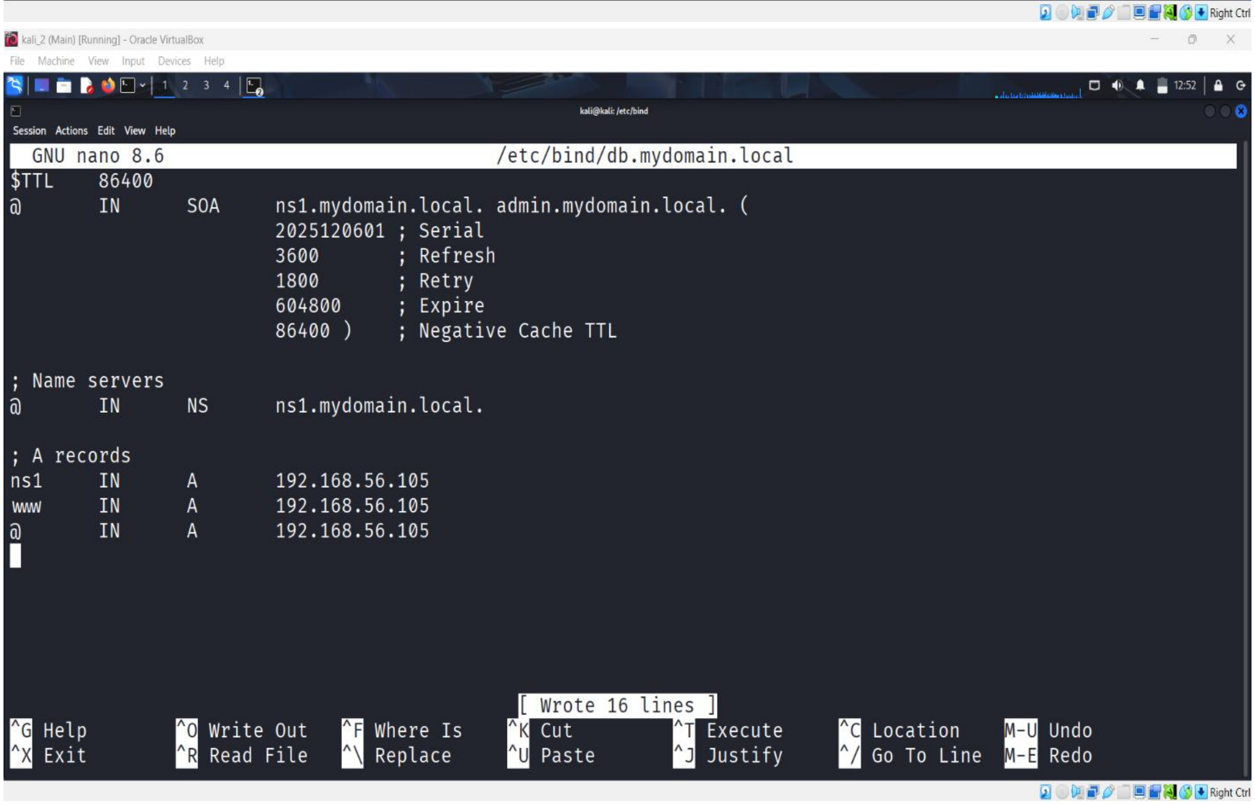
nslookup www.mydomain.local

dig ftp.mydomain.local

Screenshots:




kali_2 (Main) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: /etc/bind
Session Actions Edit View Help
GNU nano 8.6 /etc/bind/named.conf.local
//
// Do any local configuration here
//
zone "mydomain.local" {
type master;
file "/etc/bind/db.mydomain.local";
};
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo


kali_2 (Main) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
kali@kali: /etc/bind
Session Actions Edit View Help
GNU nano 8.6 /etc/bind/db.mydomain.local
$TTL 86400
@ IN SOA ns1.mydomain.local. admin.mydomain.local. (
2025120601 ; Serial
3600 ; Refresh
1800 ; Retry
604800 ; Expire
86400) ; Negative Cache TTL
; Name servers
@ IN NS ns1.mydomain.local.
; A records
ns1 IN A 192.168.56.105
www IN A 192.168.56.105
@ IN A 192.168.56.105
[Wrote 16 lines]
^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo

```
┌──(kali㉿kali)-[~]
└─$ nslookup www.mydomain.local 192.168.56.105

Server:         192.168.56.105
Address:        192.168.56.105#53

Name:   www.mydomain.local
Address: 192.168.56.105
```

kali_1 (ProjectAC) [Running] - Oracle VirtualBox

File Machine View Input Devices Help

kali@kali: ~

Session Actions Edit View Help

```
┌──(kali㉿kali)-[~]
└─$ dig www.mydomain.local


; <<>> DiG 9.20.15-2-Debian <<>> www.mydomain.local
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 65471
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 8be8bd55815d1a310100000069346f0bea787efeff88199c (good)
;; QUESTION SECTION:
;www.mydomain.local.            IN      A

;; ANSWER SECTION:
www.mydomain.local.     604800  IN      A       192.168.56.102

;; Query time: 0 msec
;; SERVER: 192.168.56.102#53(192.168.56.102) (UDP)
;; WHEN: Sat Dec 06 12:59:39 EST 2025
;; MSG SIZE  rcvd: 91


┌──(kali㉿kali)-[~]
```

12:59

Right Ctrl

3) Mail Server (Postfix)

3.1 Installation

sudo apt install postfix mailutils

Select: **Local Only**

3.2 Configuration File

/etc/postfix/main.cf

Set:

myhostname = mail.mydomain.local

mydestination = $myhostname, localhost.localdomain, localhost

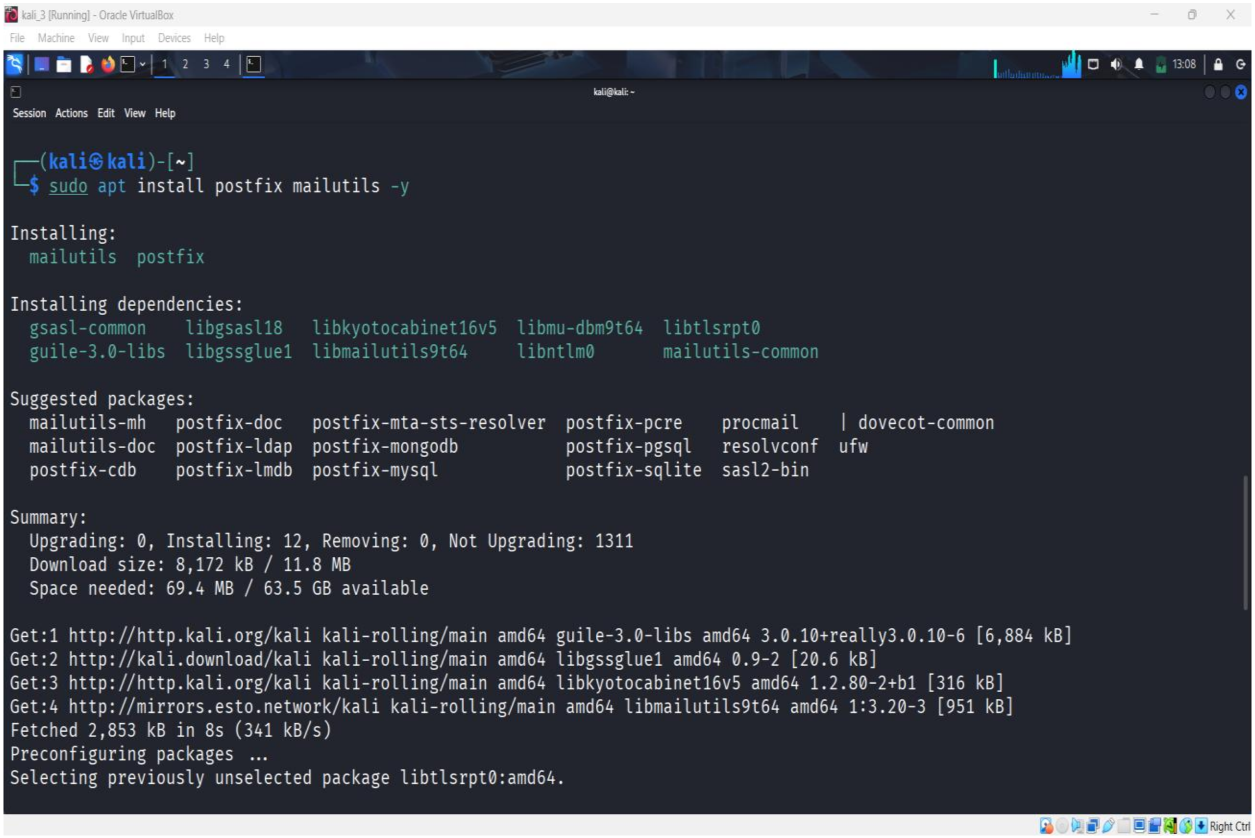
3.3 Send Test Mail

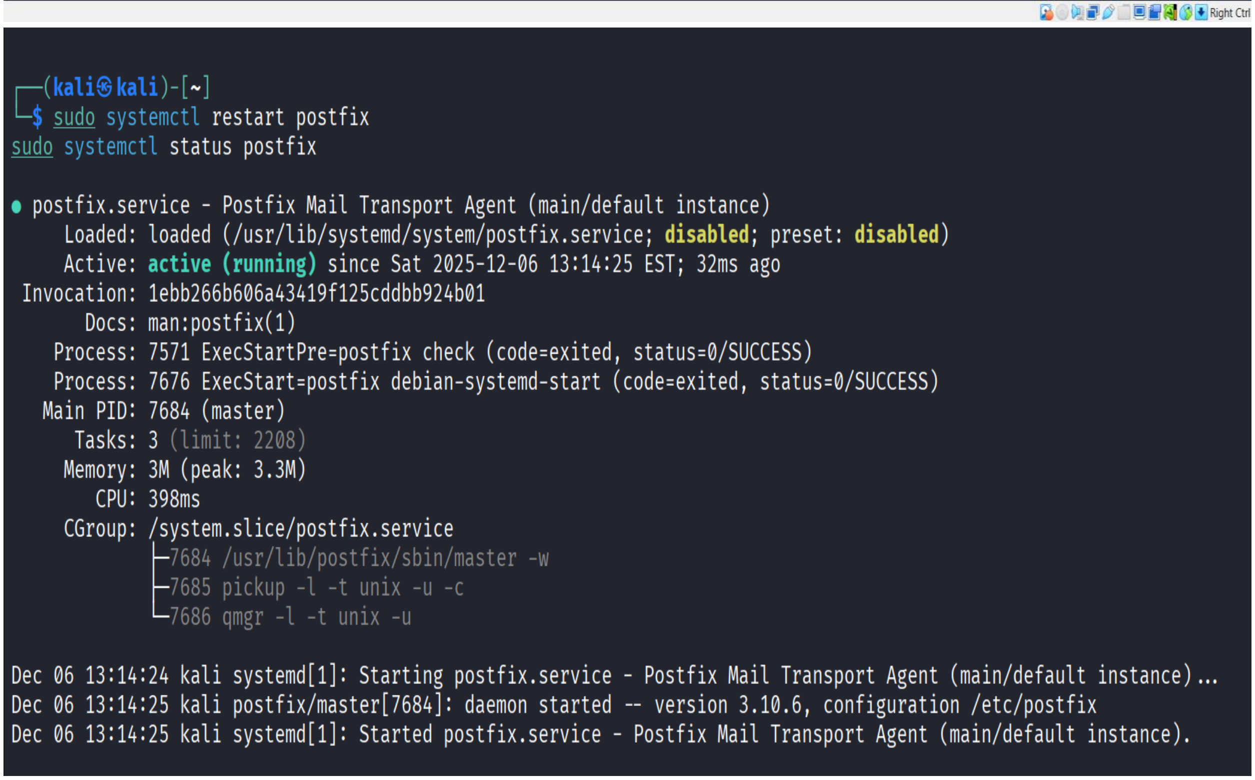
echo "Hello Maildir" | mail -s "Test Mail"user1@mydomain.local

3.4 Logs

sudo tail -f /var/log/mail.log

Screenshots





```
┌──(kali㉿kali)-[~]
└─$ echo "Hello Maildir" | mail -s "Test Mail" user1@mydomain.local
```

```
┌──(kali㉿kali)-[~]
└─$ su - user1
Password:
┌──(user1㉿kali)-[~]
└─$ cat ~/Maildir/new/*
Return-Path: <kali@kali>
X-Original-To: user1@mydomain.local
Delivered-To: user1@mydomain.local
Received: by kali.mydomain.local (Postfix, from userid 1000)
        id 2B5192C36EC; Sat, 06 Dec 2025 13:14:34 -0500 (EST)
Subject: Test Mail
To: <user1@mydomain.local>
User-Agent: mail (GNU Mailutils 3.20)
Date: Sat,  6 Dec 2025 13:14:34 -0500
Message-Id: <20251206181434.2B5192C36EC@kali.mydomain.local>
```

4) FTP Server (vsftpd)

4.1 Installation

sudo apt install vsftpd

4.2 Important Configurations

Edit:

/etc/vsftpd.conf

Set:

anonymous_enable=NO

local_enable=YES

write_enable=YES

chroot_local_user=YES

Restart:

sudo systemctl restart vsftpd

4.3 FTP Login Test

ftp 192.168.56.101

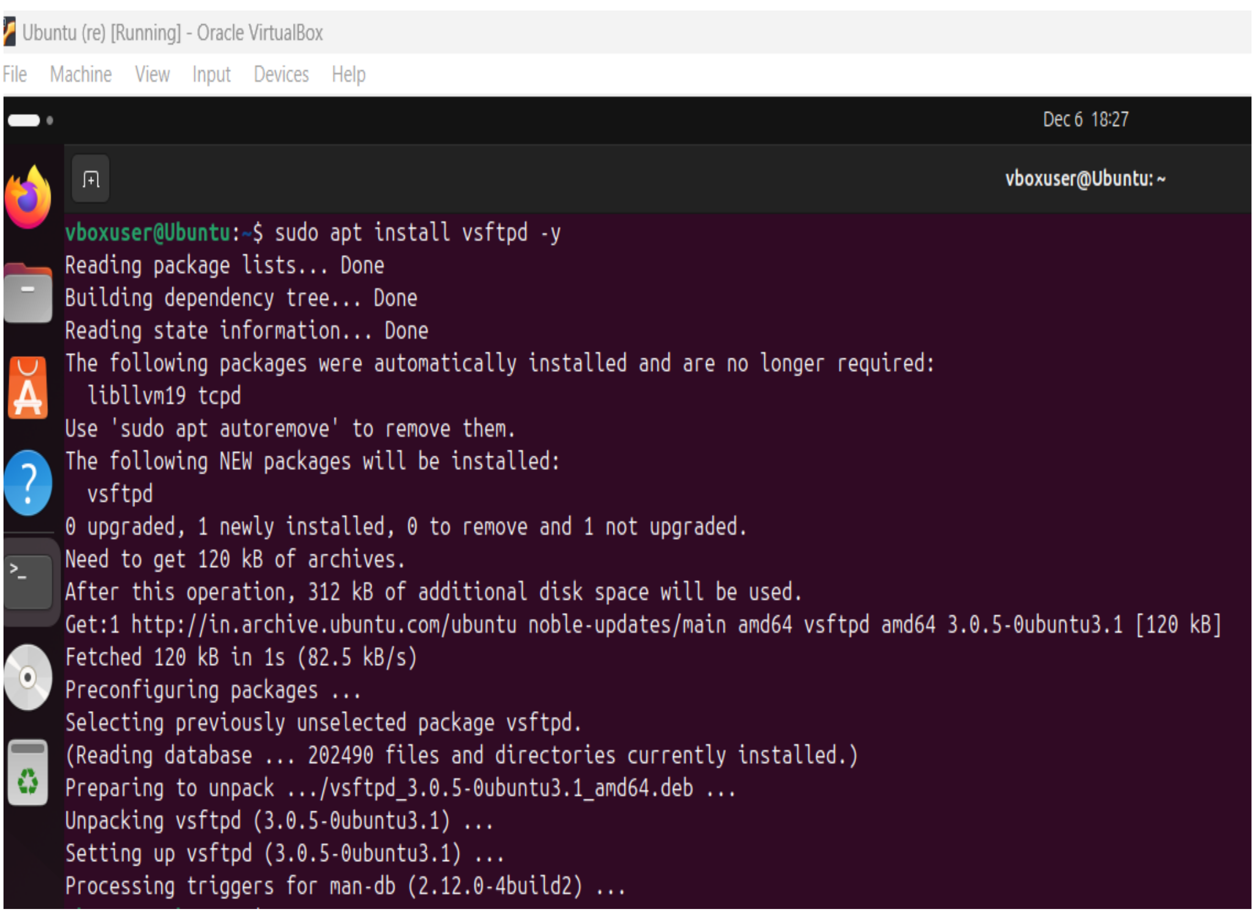
4.4 Upload / Download Test

Commands:

put file.txt

get file.txt

Screenshots



```
vboxuser@Ubuntu:~$ sudo systemctl status vsftpd
● vsftpd.service - vsftpd FTP server
     Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; disabled; preset: enabled)
     Active: active (running) since Sat 2025-12-06 18:29:57 UTC; 9s ago
    Process: 4293 ExecStartPre=/bin/mkdir -p /var/run/vsftpd/empty (code=exited, status=0/SUCCESS)
   Main PID: 4295 (vsftpd)
      Tasks: 1 (limit: 3468)
     Memory: 700.0K (peak: 1.5M)
        CPU: 15ms
     CGroup: /system.slice/vsftpd.service
             └─4295 /usr/sbin/vsftpd /etc/vsftpd.conf

Dec 06 18:29:57 Ubuntu systemd[1]: Starting vsftpd.service - vsftpd FTP server...
Dec 06 18:29:57 Ubuntu systemd[1]: Started vsftpd.service - vsftpd FTP server.
```

Ubuntu (re) [Running] - Oracle VirtualBox

File Machine View Input Devices Help

Dec 6 18:29

vboxuser@Ubuntu: ~

```
  GNU nano 7.2                          /etc/vsftpd.conf
# capabilities.
#
#
# Run standalone?  vsftpd can run either from an inetd or as a standalone
# daemon started from an initscript.
listen=NO
#
# This directive enables listening on IPv6 sockets. By default, listening
# on the IPv6 "any" address (::) will accept connections from both IPv6
# and IPv4 clients. It is not necessary to listen on *both* IPv4 and IPv6
# sockets. If you want that (perhaps because you want to listen on specific
# addresses) then you must run two copies of vsftpd with two configuration
# files.
listen_ipv6=YES
#
# Allow anonymous FTP? (Disabled by default).
anonymous_enable=NO
#
# Uncomment this to allow local users to log in.
local_enable=YES
#
# Uncomment this to enable any form of FTP write command.
write_enable=YES
#
# Default umask for local users is 077. You may wish to change this to 022,
# if your users expect that (022 is used by most other ftpd's)
local_umask=022
#
# Uncomment this to allow the anonymous FTP user to upload files. This only
# has an effect if the above global write enable is activated. Also, you will
# obviously need to create a directory writable by the FTP user.
#anon_upload_enable=YES
#
# Uncomment this if you want the anonymous FTP user to be able to create

^G Help       ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark  M-] To Bracket M-Q Previous  ^B Back
^X Exit       ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy      ^Q Where Was   M-W Next      ^F Forward
```

Right Ctrl

```
┌──(kali㉿kali)-[~]
└─$ ftp 192.168.56.101
Connected to 192.168.56.101.
220 (vsFTPd 3.0.5)
Name (192.168.56.101:kali): vboxuser
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> put test.txt
local: test.txt remote: test.txt
229 Entering Extended Passive Mode (|||60784|)
150 Ok to send data.
     0        0.00 KiB/s
226 Transfer complete.
ftp> ls
229 Entering Extended Passive Mode (|||36731|)
150 Here comes the directory listing.
drwxr-xr-x    2 1000     1000         4096 Nov 08 06:11 Desktop
drwxr-xr-x    2 1000     1000         4096 Nov 08 06:11 Documents
drwxr-xr-x    2 1000     1000         4096 Nov 08 06:11 Downloads
drwxr-xr-x    2 1000     1000         4096 Nov 08 06:11 Music
drwxr-xr-x    2 1000     1000         4096 Nov 08 06:11 Pictures
drwxr-xr-x    2 1000     1000         4096 Nov 08 06:11 Public
drwxr-xr-x    2 1000     1000         4096 Nov 08 06:11 Templates
drwxr-xr-x    2 1000     1000         4096 Nov 08 06:11 Videos
drwx------    5 1000     1000         4096 Nov 19 10:24 snap
-rw-rw-r--    1 1000     1000            0 Dec 06 18:33 test.txt
226 Directory send OK.
ftp> get test.txt
local: test.txt remote: test.txt
229 Entering Extended Passive Mode (|||9098|)
150 Opening BINARY mode data connection for test.txt (0 bytes).
     0        0.00 KiB/s
226 Transfer complete.
ftp> 
```

Security Hardening Applied

✓ UFW Firewall

- Allowed required ports: **22, 80, 443, 53, 25, 21**
- Denied everything else:

sudo ufw default deny incoming

sudo ufw default allow outgoing

```
┌──(kali㉿kali)-[~]
└─$ sudo apt install ufw -y
sudo ufw allow 22
sudo ufw allow 80
sudo ufw allow 443
sudo ufw allow 53
sudo ufw allow 21
sudo ufw enable
sudo ufw status

ufw is already the newest version (0.36.2-9).
The following packages were automatically installed and are no longer required:
  libgpgme45  python3-gpg  samba-ad-dc  samba-ad-provision  samba-dsdb-modules
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1156
Rule added
Rule added (v6)
Rule updated
Rule updated (v6)
Rule added
Rule added (v6)
Rule added
Rule added (v6)
Rule added
Rule added (v6)
Firewall is active and enabled on system startup
Status: active

To                         Action      From
--                         ------      ----
80                         ALLOW       Anywhere
22                         ALLOW       Anywhere
443                        ALLOW       Anywhere
53                         ALLOW       Anywhere
21                         ALLOW       Anywhere
80 (v6)                    ALLOW       Anywhere (v6)
22 (v6)                    ALLOW       Anywhere (v6)
443 (v6)                   ALLOW       Anywhere (v6)
53 (v6)                    ALLOW       Anywhere (v6)
```

Fail2ban

Installation:

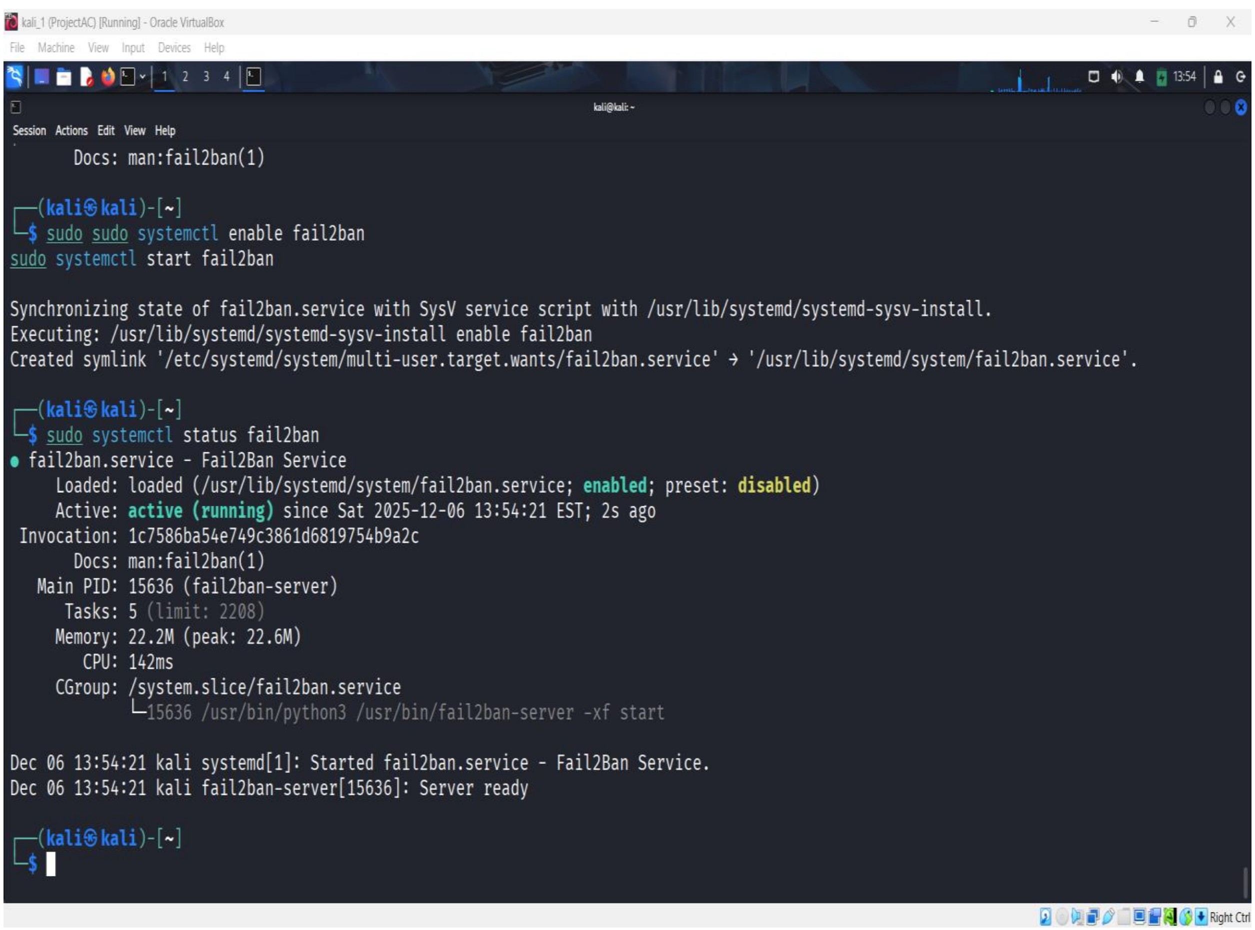
sudo apt install fail2ban

Configured jail:

/etc/fail2ban/jail.local

Enabled:

sudo systemctl enable --now fail2ban



SSH Hardening

File:

/etc/ssh/sshd_config

Set:

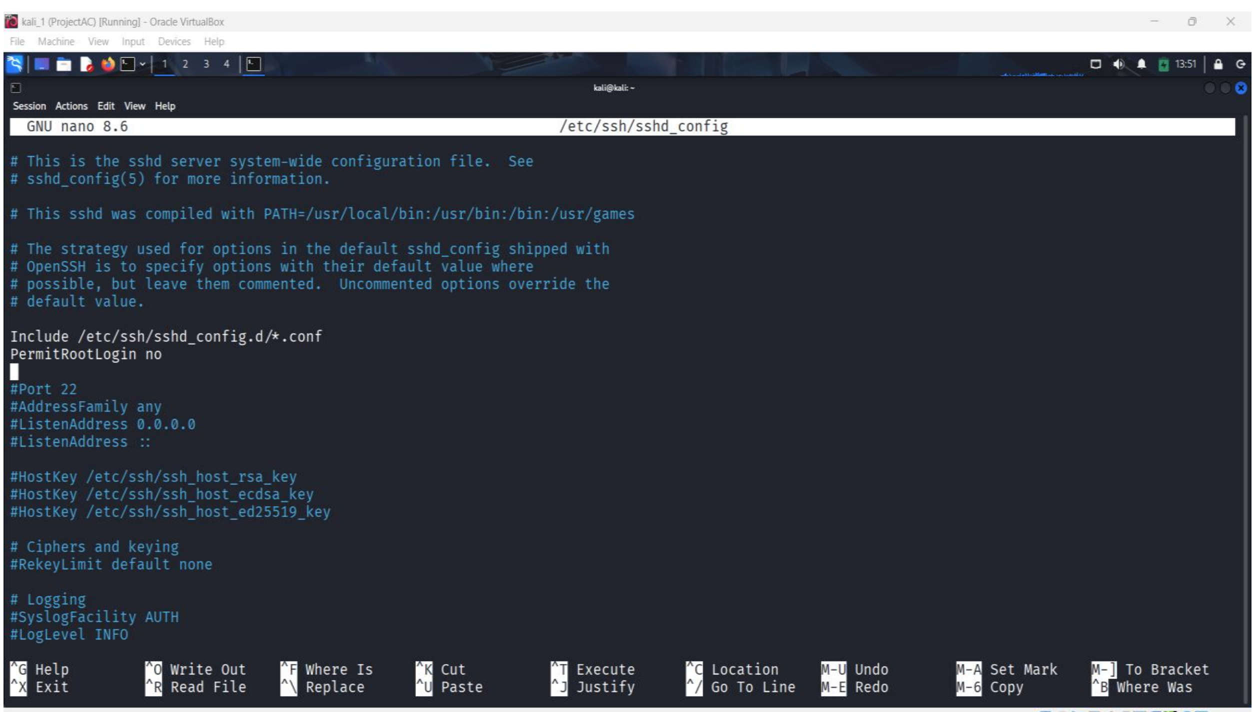
PermitRootLogin no

PasswordAuthentication yes/no

AllowUsers youruser

Restart SSH:

sudo systemctl restart ssh



FTP Hardening

- **Disabled Anonymous Login**
- **chroot jail enabled**
- **Strong password rules**
- **UFW firewall restricted**

Challenges Faced & Solutions

1. DNS resolution not working

Reason: Misconfigured zone file
Fix: Corrected A records and SOA entry

2. UFW blocked SSH accidentally

Fix: Logged in through VirtualBox console and allowed SSH manually

3. FTP permission issues

Fix: Set correct ownership and enabled write permissions

4. Postfix mail stuck in queue

Fix: Started postfix service and fixed hostname configuration

Conclusion

This project provided hands-on experience in deploying and securing multiple network services in a virtualized environment. I learned how servers communicate, how DNS and mail routing work, how to secure communication with SSL, and how to protect services using firewall and intrusion prevention tools. The lab reflects real-world system administration practices and strengthened my understanding of cybersecurity fundamentals.

In the future, this project can be enhanced by adding:

- Intrusion Detection Systems (Snort/Suricata)
- Web Application Firewall (ModSecurity)
- Load Balancing
- Cloud deployment (AWS/Azure)

References

- Apache Documentation: https://httpd.apache.org/
- Bind9 DNS Docs: https://bind9.readthedocs.io/
- Postfix Docs: http://www.postfix.org/documentation.html
- vsftpd Docs: https://security.appspot.com/vsftpd.html
- Ubuntu Official Documentation
- Nmap Manual Pages
- Fail2Ban Documentation